

Distributed Log Analytics System (Mini Splunk) - Skills Roadmap

Target: Internship-ready project demonstrating Web Development, ML, Data Science, SQL, and Big Data skills.

1. Frontend Skills

React, JavaScript, HTML, CSS, Tailwind CSS, Chart.js, API Integration, WebSockets.

2. Backend Skills

Python, FastAPI, REST APIs, Authentication, Logging, WebSockets.

3. Database Skills

PostgreSQL, SQL Queries, Indexing, Aggregations, Joins, Database Design.

4. Big Data Skills

Apache Kafka, Apache Spark Streaming, Hadoop Ecosystem, Distributed Data Processing.

5. Machine Learning Skills

Anomaly Detection, Isolation Forest, Random Forest, XGBoost, Feature Engineering, Model Evaluation.

6. DevOps & Deployment

Docker, Git, GitHub, CI/CD Basics, Azure or AWS Deployment.

Project Features

Log ingestion, search engine, real-time monitoring, anomaly detection, AI log summarization, analytics dashboard.

30-Day Development Plan

Week 1: React Dashboard + FastAPI + PostgreSQL

Week 2: Log Ingestion + Search + Charts

Week 3: Kafka + Spark Streaming + Real-Time Updates

Week 4: ML Anomaly Detection + Docker + Deployment + Documentation